BUGRAHAN GURESCI

SENIOR MECHANICAL ENGINEERING STUDENT

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EDUCATION

Bogazici University
2020-2026
Mechanical Engineering
(2.61 / 4.0)

Nilufer Borsa Istanbul Science High School
2016-2020
98.98 / 100

Ali Kutuk Middle School
2012-2026
Graduated with the highest GPA

RELEVANT SKILLS

- MATLAB
- Python
- Solidworks
- Microcontrollers & Embedded systems
- MS Office
- JavaScript

LANGUAGES

- English *Advanced*
- Turkish *Native*

CERTIFICATES

- International Ride Operator Certification (iROC)

SUMMARY

I am eager to take on responsibilities after a short adjustment period in unfamiliar fields. I approach challenges with a balanced mindset—prioritizing reliability while embracing calculated risks when necessary.

WORK EXPERIENCE

LABORATORY ASSISTANT: SOLID-STATE ELECTROLYTE LABORATORY

BOGAZICI University, Istanbul
November 2025-

- Synthesized and characterized high-purity NASICON ceramic electrolyte via the Solid-State Reaction (SSR) route, managing stoichiometric precursor mixing and multi-stage ball milling with phase verification through X-Ray Diffraction (XRD) .

PROJECT INTERN, SALES (EAST EUROPE)

SISECAM Headquarters, Istanbul
January 2026-

- Supported the Eastern Europe sales team by identifying new market opportunities, managing client communications, and coordinating sample distributions to facilitate price negotiations and sales growth

PRODUCTION INTERN, FIRST STEP

SISECAM Glass Packaging Factory, Bursa
August 2025- September 2025 (2 Months)

- The time study project is prepared, the furnace and its structure are examined, process optimization suggestions are presented, and the glass packaging production process is observed.

SYSTEM INTEGRATION INTERN, STAGE

ROKETSAN, Ankara
June 2025- July 2025 (2 Months)

- Analyzed the cooling-loop performance for the launcher system, calculated head losses and system characteristic curve, and pump performance curve at different rpms, converted into MATLAB code.

RIDE OPERATOR

Cedar Point, Ohio, USA
June 2023- October 2023 (4 Months)

- Conducted daily equipment tests before operation, verified safety restraints, operated the ride from control room, collaborated with the team to manage the technical malfunctions.

YOUTH EXCHANGE, RESOLVE: PLAY 4 PEACE

Erasmus+, London, UK
30th October-5th November 2022

- Provided a safe space for intercultural dialogue, fostering tolerance and mutual understanding among young through presentations and discuss as a group.

PROJECTS

SUBMARINE

- Developed the buoyancy system of a remotely controlled submarine with three-axis movement capability, contributed to coding and mechanical design, ensuring functional integration of motor drivers, communication modules, servo motors, etc.

HEAT RECOVERY SYSTEM

- Designed the heat recovery system for residential fireplaces, analyzed the heat transfer, pressure drop, and cost calculations.
- Implemented into MATLAB code to optimize thousands of configurations

AUTOMATED MICROMETER-ADJUSTABLE DR BLADE TAPE CASTING MACHINE WITH HEATED VACUUM CHUCK AND COOLING SYSTEM (FINAL YEAR PROJECT)

- Designed and manufactured a custom automated tape casting machine for our laboratory work NASICON solid-state electrolyte processing, integrating complete mechanical assembly and PDS developed in SolidWorks .
- Programmed a NEMA 23 stepper motor and TB6600 driver to regulate precise traverse speeds (adjustable 0-100mm/s) via a T8 lead screw and belt-pulley transmission system
- Engineered an aluminum vacuum chuck via Fast Hole EDM Drilling and CNC Lathe to ensure uniform, substrate fixation, integrated a closed-loop PID thermal control system to precisely manage solvent evaporation
- Performed technical analyses including steady-state heat transfer simulations, thermal expansion modeling at 120°C, and fluid dynamic drag force vs. viscosity calculations.